

ECON 803 Microeconomic Theory I

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Question-only Practice Book - Fall 2026

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How to Use This Practice Book

This volume collects the complete student-facing problem program for ECON 803. It contains 66 chapter problems and 10 additional-practice problems.

Part I problems are compact benchmarks intended for derivation and discussion by hand. Part II problems develop proofs, applications, and one named use case for each chapter. Labels identify Core, Proof, and Applied work; they are not difficulty rankings.

Before using Codex, Python, or Lean, record four lines: Seed identifies the hand result being extended; Fixed records the assumptions that remain unchanged; Scaled names the single enlarged dimension or strengthened claim; Risk states the new failure mode. The resulting artifact must reproduce the hand case and include a boundary or deliberately failing test.

A successful computation or compilation verifies only the statement encoded in the artifact. Students remain responsible for the economic assumptions, the interpretation, and the distinction between numerical evidence and proof.

Fall 2026 Weekly Practice Map

This map organizes the non-case chapter problems into thirteen weekly sets. Chapter use cases remain a separate case-study stream. Reserve problems are optional additional practice. The stable identifiers below refer to the full questions later in this book.

PS01: Utility maximization and demand

Class meeting: Sep. 11, 2026. Due: Sep. 18, 2026.

Required: 1.1, 1.2, 1.4. Formal or applied extension: 1.3, 1.6. Optional reserve: None.

PS02: Duality and welfare

Class meeting: Sep. 18, 2026. Due: Sep. 25, 2026.

Required: 2.1, 2.3, 2.5. Formal or applied extension: 2.2, 2.4. Optional reserve: None.

PS03: Revealed preference and recoverability

Class meeting: Sep. 25, 2026. Due: Oct. 2, 2026.

Required: 3.1, 3.3, 3.6. Formal or applied extension: 3.2, 3.5. Optional reserve: R001.

PS04: Abstract choice and order foundations

Class meeting: Oct. 2, 2026. Due: Oct. 9, 2026.

Required: 4.1, 4.3, A.1. Formal or applied extension: A.2. Optional reserve: R002, R004.

PS05: Expected utility and risk

Class meeting: Oct. 9, 2026. Due: Oct. 16, 2026.

Required: 5.1, 5.2. Formal or applied extension: 5.3. Optional reserve: None.

PS06: Stochastic choice and attention

Class meeting: Oct. 16, 2026. Due: Oct. 23, 2026.

Required: 6.1, 6.3, 7.1. Formal or applied extension: 7.2. Optional reserve: R003.

PS07: Aggregation

Class meeting: Oct. 23, 2026. Due: Oct. 30, 2026.

Required: 8.1, 8.2. Formal or applied extension: None. Optional reserve: None.

PS08: Production and cost

Class meeting: Oct. 30, 2026. Due: Nov. 6, 2026.

Required: 9.1, 9.3. Formal or applied extension: 9.2. Optional reserve: R005.

PS09: Competitive markets and exchange

Class meeting: Nov. 6, 2026. Due: Nov. 13, 2026.

Required: 10.1, 10.2, 11.1. Formal or applied extension: 10.3. Optional reserve: R006.

PS10: Welfare and positive equilibrium

Class meeting: Nov. 13, 2026. Due: Nov. 20, 2026.

Required: 11.2, 12.1, 12.2. Formal or applied extension: 11.3, 12.3. Optional reserve: R008, R009.

PS11: Empirical equilibrium and asset markets

Class meeting: Nov. 20, 2026. Due: Nov. 27, 2026.

Required: 13.1, 14.1, 14.2. Formal or applied extension: 13.2. Optional reserve: R007, R010.

PS12: Matching

Class meeting: Nov. 27, 2026. Due: Dec. 4, 2026.

Required: 15.1, 15.2, A.3. Formal or applied extension: None. Optional reserve: None.

PS13: Computable equilibrium

Class meeting: Dec. 4, 2026. Due: Dec. 7, 2026.

Required: 16.1, 16.2. Formal or applied extension: None. Optional reserve: None.

Part I

The Classical Rational Consumer

Chapter 1

Utility Maximization and Consumer Demand

Part I: By hand

Problem 1.1 [Core]. Budget scaling and Cobb-Douglas demand.

Let $X = \mathbb{R}_+^2$ and $B(p, w) = \{x \in X : p \cdot x \leq w\}$, where $p \gg 0$ and $w > 0$.

(a) Prove that $B(\lambda p, \lambda w) = B(p, w)$ for every $\lambda > 0$. Explain why every demand correspondence generated by preference maximization is homogeneous of degree zero in (p, w) .

(b) Suppose preferences are locally nonsatiated. Prove that every chosen bundle satisfies Walras' law, $p \cdot x = w$.

(c) For $u(x) = x_1^\alpha x_2^{1-\alpha}$ with $0 < \alpha < 1$, derive Marshallian demand and verify both properties directly.

Problem 1.2 [Core]. CES curvature, demand, and expenditure shares.

For $\rho < 1$ and $\rho \neq 0$, let $u_\rho(x) = (\frac{1}{2}x_1^\rho + \frac{1}{2}x_2^\rho)^{1/\rho}$, with the Cobb-Douglas limit used at $\rho = 0$ and the boundary extension stated in the text. Write $\sigma = 1/(1 - \rho)$.

(a) Derive demand from the first-order conditions. Explain why the optimum is interior for $\rho < 1$ and why maximizing the sum of powers is incorrect when $\rho < 0$.

(b) At $p = (1, 4)$ and $w = 120$, calculate quantities and expenditure shares for $\sigma \in \{1/2, 1, 2\}$. Check budget exhaustion.

(c) Raise only the second price to 9. Recalculate the three demands and explain why a smaller quantity can have a larger expenditure share.

(d) Derive the own-price, cross-price, and wealth elasticities. Distinguish the elasticity of a quantity ratio from an individual demand elasticity.

Problem 1.4 [Core]. Substitutes, complements, and boundary choices.

Consider separately $u^A(x) = 2x_1 + x_2$ and $u^B(x) = \min\{x_1/3, x_2\}$ on $\mathbb{R}_+^2$, at strictly positive prices and wealth.

(a) Derive the full demand correspondence for u^A , including the price ratio at which it is set-valued.

(b) Derive demand for u^B . At $p = (2, 3)$ and $w = 36$, compute both models' choices.

(c) Explain why interior differentiable tangency conditions do not characterize these optima.

(d) Obtain the ratio $x_1 = 3x_2$ as a CES limit. Explain why holding the usual taste weights fixed does not preserve unequal Leontief proportions.

Part II: Formal and applied extensions

Problem 1.3 [Applied]. CES calibration and observational equivalence.

At prices $p^0 = (1, 2, 4)$, a household with wealth $w^0 = 120$ chooses $x^0 = (30, 25, 10)$. Consider Chapter 1's CES representation with positive normalized weights.

- (a) For a specified $\sigma > 0$, derive the weights reproducing this observation.
- (b) Calibrate the weights for $\sigma = 1/2$ and $\sigma = 2$. For each calibration predict demand at $p^1 = (2, 2, 4)$ and wealth 120.
- (c) Explain why fitting one observation does not identify substitution. Describe additional observations that could distinguish the models.

Problem 1.6 [Applied]. KKT regimes and the quantity--quality tradeoff.

Unified Growth Theory links the demographic transition to a shift from child quantity toward education and child quality. This exercise isolates a stylized household block from Galor (2005); it is not the complete dynamic growth model. A parent with potential income $y > \bar{c}$ chooses consumption $c > 0$, fertility $n > 0$, and education time per child $e \geq 0$ to maximize $(1 - \gamma) \log c + \gamma \log[nh_a(e)]$, where $h_a(e) = 1 + ae/(\tau + e)$, $0 < \gamma < 1$, $\tau > 0$ is baseline time per child, $a > 0$ indexes the return to education, and $\bar{c} > 0$ is subsistence consumption. The resource and subsistence constraints are $c + yn(\tau + e) \leq y$ and $c \geq \bar{c}$. The function h_a is a closed-form teaching specialization of Galor's general child-human-capital technology; the normalization makes one unit of e one unit of parental time.

Part I. By hand

- (a) Write the complete KKT system in (c, n, e) , including feasibility, multiplier signs, and complementary slackness. Prove that the resource constraint binds.
- (b) Define total child-rearing time $t = n(\tau + e)$. Eliminate c and n , state the feasible set for (t, e) , and show that the transformed objective is additively separable in t and e . Write its KKT system. Explain why an inequality can be active with a zero multiplier at a threshold.
- (c) Solve globally for (t^*, e^*) , then recover (c^*, n^*) . Divide the answer into all parameter regions, including the equality cases. A stationary-point calculation alone is not sufficient. Hint: determine the sign of $h'_a(e)(\tau + e) - h_a(e)$.
- (d) Let $\tau = 1/4$, $\gamma = 1/2$, and $\bar{c} = 4$. Calculate $(c^*, n^*, e^*, h_a(e^*))$ for every combination $a \in \{1/2, 3\}$ and $y \in \{6, 12\}$. Identify the active inequalities, calculate their transformed-problem multipliers, and verify feasibility and complementary slackness.
- (e) Explain the income mechanism while subsistence binds, the quantity--quality response when a rises, and which dynamic elements of Unified Growth Theory this static exercise deliberately leaves out.

Part II. Verified scale-up with Codex and Python

- (f) Record four lines before coding: Seed: the four hand calculations; Fixed: preferences, h_a , continuous fertility, and $(\tau, \gamma, \bar{c})$; Scaled: only the number of (a, y) configurations; Risk: threshold

misclassification, invalid parameter domains, or mistaking finite computation for proof.

(g) Create a file named `work/ugt_kkt.py`. On a deterministic rectangular grid containing $a = 1$ and $y = \bar{c}/(1 - \gamma)$, compute the closed-form candidate and the transformed-problem KKT multipliers. Reject $a \leq 0$, $\tau \leq 0$, $\bar{c} \leq 0$, $\gamma \notin (0, 1)$, and $y \leq \bar{c}$.

(h) At every grid point, check the resource identity, subsistence feasibility, nonnegativity, stationarity, complementary slackness, and agreement with an independently implemented bounded numerical maximization. State the numerical tolerances. Plot e^* and n^* and mark the two analytic threshold lines.

(i) Include tests immediately below, at, and above both thresholds. Report separately what the hand argument proves for every admissible parameter vector, what the program checks only on its finite grid, and what remains an empirical or dynamic modeling claim.

Household demand across subsistence and discretionary regimes

Chapter use case. Seed: Problems 1.1, 1.4. Fixed: all unlisted primitives and assumptions. Scaled: wealth from two hand calculations to a grid, with preferences and prices fixed. Tool: Codex + Python. Verification: Reproduce both hand wealth levels; classify infeasible, boundary, and interior regimes separately for each model; check feasibility, budget exhaustion, and continuity at every valid threshold.

Problem 1.5 [Applied]. Subsistence and quasilinear corners.

The first of two consumers has $u(x) = (x_1 - 1)^{1/3}(x_2 - 2)^{2/3}$ on $x \geq (1, 2)$. The second has $u(x) = 4\sqrt{x_1} + x_2$ on $\mathbb{R}_+^2$.

(a) Derive the first consumer's demand above subsistence expenditure. State what happens at and below that threshold.

(b) At $p = (2, 1)$, compute the first consumer's demands and shares at wealth 10 and 22. Distinguish homotheticity in discretionary consumption from homotheticity in total consumption.

(c) Normalize the second consumer's numeraire price to one and write $r = p_1$. Derive demand for all $r, w > 0$, including the zero-numeraire region.

(d) Identify where a zero wealth effect for the non-numeraire good is valid. Compare this restriction with the Stone--Geary model.

Chapter 2

Duality, Comparative Statics, and Welfare

Part I: By hand

Problem 2.1 [Core]. Cobb-Douglas duality.

For $u(x) = x_1^\alpha x_2^{1-\alpha}$ with $0 < \alpha < 1$, $p \gg 0$, $w > 0$, and $\bar{u} > 0$, derive the Marshallian demand $x(p, w)$, indirect utility $v(p, w)$, Hicksian demand $h(p, \bar{u})$, and expenditure function $e(p, \bar{u})$.

- (a) Verify Walras' law and the identity $e(p, v(p, w)) = w$.
- (b) Verify $h(p, v(p, w)) = x(p, w)$.
- (c) Verify Roy's identity and Shephard's lemma.

Problem 2.5 [Core]. Translog shares and the regular price region.

Consider a homothetic translog index with $\alpha_0 = 0$, $\alpha = (2/5, 3/5)$, and $\Gamma = -\frac{1}{8} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$.

- (a) Derive shares and demands as functions of relative prices and wealth. Find the open interval of p_1/p_2 for which both shares are positive.
- (b) Compute $H = \Gamma + \mathbf{b}\mathbf{b}^\top - \text{diag}(\mathbf{b})$. Prove negative semidefiniteness throughout that interval.
- (c) At $p = (2, 1)$ and $w = 100$, calculate demand, Marshallian elasticities, and compensated elasticities. Verify adding up and homogeneity.
- (d) Explain why setting a negative predicted share to zero while retaining the other formula does not generally give a valid extension.

Part II: Formal and applied extensions

Problem 2.2 [Proof]. Differential restrictions on demand.

Let $x : \mathbb{R}_{++}^L \times \mathbb{R}_{++} \rightarrow \mathbb{R}_+^L$ be continuously differentiable and satisfy Walras' law.

- (a) Differentiate Walras' law with respect to wealth and prices.
- (b) Define the Slutsky matrix by $S(p, w) = D_p x(p, w) + D_w x(p, w)x(p, w)^\top$. Prove that $p^\top S(p, w) = 0$.
- (c) Explain which additional restrictions on S follow from utility maximization and which of them do not follow from Walras' law alone.

Problem 2.3 [Proof]. The Slutsky matrix for Cobb-Douglas demand.

Using the Marshallian demand from Problem 2.1, compute $S(p, w) = D_p x + D_w x x^\top$.

- (a) Show that S is symmetric.
- (b) Show directly that $z^\top S z \leq 0$ for every $z \in \mathbb{R}^2$.
- (c) Find the null space of S and interpret it using homogeneity.

Problem 2.4 [Proof]. CES duality and compensated substitution.

Use Chapter 1's CES utility, with positive normalized weights and $\sigma > 0$. For $\sigma \neq 1$, put $D(p) := \sum_k a_k^\sigma p_k^{1-\sigma}$.

- (a) Derive $\mathcal{P}(p)$, expenditure, and Hicksian demand. Treat $\sigma = 1$ separately and retain the constants implied by the utility normalization.
- (b) Show that the Slutsky matrix has off-diagonal entries $S_{\ell k} = \sigma x_\ell x_k / w$ and diagonal entries $S_{\ell \ell} = -\sigma x_\ell (1 - b_\ell) / p_\ell$. Prove symmetry and negative semidefiniteness using a weighted-variance argument. Identify its null space for strictly positive demand.
- (c) Explain how compensated cross-price effects can be positive when uncompensated cross-price effects are negative.
- (d) For equal weights, $\sigma = 2$, initial prices $(1, 3)$, wealth 90, and final prices $(2, 3)$, calculate compensation to retain initial utility. Decompose the first quantity's change into a compensated change and the remaining wealth effect.

Local CES and translog welfare approximation

Chapter use case. Seed: Problems 2.4, 2.5. Fixed: all unlisted primitives and assumptions. Scaled: log relative-price distance from one reference price. Tool: Codex + Python. Verification: Match the CES value, gradient, and Hessian at the reference price; enforce positive shares and negative semidefiniteness on the translog regular region; report welfare comparisons only on that region.

Problem 2.6 [Applied]. A translog approximation to CES demand.

Fix a two-good CES model, a positive reference price vector p^0 , and substitution elasticity $\sigma > 0$. Write b_1^0, b_2^0 for the reference shares and use normalized prices $r_\ell = p_\ell / p_\ell^0$.

- (a) Compute the gradient and Hessian of $\log \mathcal{P}_{\text{CES}}$ with respect to log prices at p^0 .
- (b) Choose translog coefficients in $\log r$ so the log price index agrees with CES through second order at p^0 .
- (c) Verify the restrictions for homogeneity. Show that demand levels and local price responses agree at the reference observation.
- (d) Explain why predictions can disagree for a large price change. State the regularity checks needed before making welfare comparisons.

Chapter 3

Revealed Preference and Recoverability

Part I: By hand

Problem 3.1 [Core]. A two-observation WARP test.

A consumer chooses $x^1 = (2, 2)$ at $(p^1, w^1) = ((2, 1), 6)$ and $x^2 = (1, 5/2)$ at $(p^2, w^2) = ((1, 2), 6)$.

- (a) Verify that each observed bundle exhausts its budget.
- (b) Determine whether each bundle was affordable at the other observation.
- (c) Does the dataset satisfy WARP for single-valued demand? Identify the two direct revealed-preference comparisons that determine your answer.

Problem 3.2 [Core]. Recovering a quasilinear utility.

Normalize $p_2 = 1$ and suppose an interior demand system satisfies $x_1(p_1, w) = \beta - \alpha p_1$, where $\alpha, \beta > 0$, and Walras' law.

- (a) Derive $x_2(p_1, w)$ and state the domain on which both demands are nonnegative.
- (b) Compute the normalized scalar Slutsky term for the non-numeraire demand.
- (c) Recover a quasilinear utility representation $u(x_1, x_2) = g(x_1) + x_2$ that generates the demand on the interior domain. Distinguish general ordinal uniqueness up to an increasing transformation from the narrower normalization freedom after the coefficient on x_2 is fixed at one.

Problem 3.3 [Core]. Afriat inequalities by hand.

Consider two observations: $p^1 = (1, 1)$, $x^1 = (2, 1)$ and $p^2 = (2, 1)$, $x^2 = (1, 1)$.

- (a) Compute observed expenditures and all direct revealed-preference comparisons.
- (b) Check GARP.
- (c) Verify that $U^1 = 1$, $U^2 = 0$, and $\lambda^1 = \lambda^2 = 1$ satisfy the Afriat inequalities.
- (d) Write the corresponding piecewise-linear Afriat utility and verify that it assigns the required utility levels at the observations.

Problem 3.5 [Core]. Classical utility as singleton CMU.

Let $u : X \rightarrow \mathbb{R}$ be continuous, let $\Omega_u = \{\{u\}\}$, and define $r_{\Omega_u}(x, y) = \max_{U \in \Omega_u} \min_{v \in U} [v(x) - v(y)]$.

- (a) Prove that $r_{\Omega_u}(x, y) = u(x) - u(y)$.
- (b) Prove that $x_{r_{\Omega_u}}(p, w) = \arg \max_{x \in B(p, w)} u(x)$.
- (c) Explain precisely which conclusions require continuity, strict monotonicity, or concavity of u , and which do not.

Part II: Formal and applied extensions

Problem 3.6 [Proof]. WGARP without GARP.

Consider prices $p^1 = (4, 1, 5)$, $p^2 = (5, 4, 1)$, $p^3 = (1, 5, 4)$ and choices $x^1 = (4, 1, 1)$, $x^2 = (1, 4, 1)$, $x^3 = (1, 1, 4)$.

- (a) Compute the 3×3 matrix with (t, s) entry $p^t \cdot x^s$.
- (b) Show that the data satisfy WGARP and WARP but violate GARP. Identify the revealed three-cycle and explain why no pair forms a reversal.
- (c) Suppose an asymmetric, strictly increasing preference function r rationalizes two choices $x \in x_r(p, w)$ and $x' \in x_r(p', p' \cdot x)$. Prove the compensated law of demand $(p' - p) \cdot (x' - x) \leq 0$. State where nonemptiness of demand is used.

A counterfactual demand support set

Chapter use case. Seed: Problems 3.1, 3.3. Fixed: all unlisted primitives and assumptions. Scaled: the number of observed budgets from two to T , with two goods and the counterfactual budget fixed. Tool: Codex + Python; Lean for candidate certificates. Verification: Recover the hand support interval and verify that added observations can only shrink it. Lean may check GARP or Afriat certificates for candidates; it does not yet certify the full support-set characterization.

Problem 3.4 [Applied]. Demand forecasting with Varian's support set.

A finite dataset $\mathcal{O} = \{(p^t, x^t)\}_{t \in \mathbb{T}}$ satisfies GARP. Fix a counterfactual budget (p^0, w^0) with $p^0 \gg 0$, and define $\mathcal{S}_{\mathcal{O}}(p^0, w^0) = \{x^0 : p^0 \cdot x^0 = w^0, \mathcal{O} \cup \{(p^0, x^0)\} \text{ satisfies GARP}\}$.

- (a) State a revealed-preference condition under which the augmented data imply $x^t \succeq^R x^0$.
- (b) Derive the associated linear restriction on $p^0 \cdot x^0$.
- (c) Explain why the restrictions in parts (a)–(b) are necessary but must be combined with every revealed-preference path into and out of the candidate observation to characterize $\mathcal{S}_{\mathcal{O}}(p^0, w^0)$. Explain why this set gives bounds rather than point identification.

Chapter 4

Choice Without Budget Geometry

Part I: By hand

Problem 4.1 [Core]. Incomplete preferences and multiple utilities.

Let $X = \{a, b, c\}$ and let R contain all reflexive pairs and the comparison aRb , but no other comparison between distinct alternatives.

- (a) Show that R is a preorder but is incomplete.
- (b) Prove that no real-valued utility function can represent R in the sense xRy if and only if $u(x) \geq u(y)$.
- (c) Construct two utility functions u_1, u_2 such that xRy if and only if $u_k(x) \geq u_k(y)$ for both $k = 1, 2$.

Part II: Formal and applied extensions

Problem 4.3 [Proof]. Maximization and WGARP.

Let R be complete and transitive on a finite set X . For each nonempty menu $A \subseteq X$, let $c(A)$ be the set of R -greatest elements of A .

- (a) Prove that $c(A)$ is nonempty.
- (b) Suppose $x \in c(A)$, $y \in c(B)$, and $x, y \in A \cap B$. Prove that xI_Ry , where xI_Ry means xRy and yRx .
- (c) Explain why this conclusion is WGARP, the appropriate weak axiom for a set-valued choice correspondence, and distinguish it from WARP for single-valued demand.

A menu-design rationality audit

Chapter use case. Seed: Problem 4.2. Fixed: all unlisted primitives and assumptions. Scaled: the number of observed menus on the fixed set $X = \{a, b, c\}$. Tool: Codex + Lean. Verification: Recover the three-menu cycle and the one-observation repair; return an exact cycle certificate or

a complete transitive rationalization. The current pairwise-WARP theorem does not exclude this three-cycle; the general checker theorem is a new target.

Problem 4.2 [Core]. Choice cycles on finite menus.

Let $X = \{a, b, c\}$ and suppose a single-valued choice function satisfies $c(\{a, b\}) = a$, $c(\{b, c\}) = b$, and $c(\{a, c\}) = c$. Say that a preference relation R rationalizes these data when the observed alternative is the unique R -greatest element of each observed menu.

- (a) Write the direct revealed-preference comparisons generated by these choices.
- (b) Show that no transitive preference relation can rationalize the data.
- (c) State the smallest change to one observation that restores acyclicity, and exhibit a complete and transitive rationalization of the revised data.

Part II

Risk, Stochastic Choice, and Attention

Chapter 5

Expected Utility and Risk

Part I: By hand

Problem 5.1 [Core]. Affine expected-utility representations.

Let $Z = \{z_1, \dots, z_n\}$ and suppose $U(\mu) = \sum_j \mu_j u(z_j)$ represents a nontrivial preference on $\Delta(Z)$.

- (a) Show directly that U is affine in lotteries.
- (b) Prove that $U' = aU + b$ with $a > 0$ represents the same preference.
- (c) Show that if two expected-utility representations agree on preferences, their Bernoulli indices differ by a positive affine transformation.

Problem 5.2 [Core]. First-order stochastic dominance.

Lotteries μ and ν on prizes $\{0, 1, 2\}$ have probabilities $\mu = (0.2, 0.3, 0.5)$ and $\nu = (0.3, 0.4, 0.3)$.

- (a) Compare their cumulative distribution functions and determine which lottery first-order stochastically dominates the other.
- (b) Prove the expected-utility ranking for every increasing Bernoulli utility by discrete summation by parts.
- (c) Explain why strict dominance need not imply a strict preference for a weakly increasing utility.

Part II: Formal and applied extensions

Problem 5.3 [Proof]. The Arrow-Pratt approximation.

Let $\varepsilon_\sigma = \sigma\xi$, where $\mathbb{E}\xi = 0$, $\mathbb{E}\xi^2 = 1$, and ξ is bounded. An agent with wealth w has utility u that is twice continuously differentiable near w , with $u'(w) > 0$. For all sufficiently small $\sigma \geq 0$, local invertibility gives a unique solution near zero to the indifference equation below. Define the monetary risk premium $\text{RP}(\sigma)$ to be that local branch, characterized by $\text{RP}(\sigma) \rightarrow 0$ as $\sigma \downarrow 0$:

$$u(w - \text{RP}(\sigma)) = \mathbb{E}[u(w + \varepsilon_\sigma)].$$

- (a) Use second-order Taylor expansions to derive the approximation

$$\text{RP}(\sigma) = \frac{1}{2}A(w)\sigma^2 + o(\sigma^2),$$

where $A(w) = -u''(w)/u'(w)$.

- (b) State the regularity and small-risk qualifications needed for this approximation.
- (c) Explain why the formula is invariant to positive affine transformations of u .

A multi-asset portfolio under risk

Chapter use case. Seed: Problems 5.3, 5.4. Fixed: all unlisted primitives and assumptions. Scaled: the number of jointly normal risky assets from one to K , retaining CARA utility and unconstrained positions. Tool: Codex + Python. Verification: Recover the one-asset formula; verify the first-order condition and concavity using the covariance matrix; report singular-covariance cases explicitly and do not add portfolio constraints silently.

Problem 5.4 [Applied]. CARA-normal portfolio demand.

Terminal wealth is $W_T = wR_f + \theta(\tilde{R} - R_f)$, where $\tilde{R} \sim N(m_R, \sigma^2)$, R_f is risk free, $\theta \in \mathbb{R}$ (borrowing and short selling are unrestricted), and utility is $u(W_T) = -\exp(-aW_T)$ with $a > 0$.

- (a) Show that maximizing expected utility is equivalent to maximizing a certainty equivalent.
- (b) Derive optimal risky investment θ^* .
- (c) Give comparative statics with respect to $m_R - R_f$, a , and σ^2 .

Chapter 6

Stochastic Choice and Random Utility

Part I: By hand

Problem 6.1 [Core]. Logit odds and IIA.

Choice probabilities satisfy $P(j \mid A) = \exp(\delta_j/\tau) / \sum_{k \in A} \exp(\delta_k/\tau)$, where $\tau > 0$ and δ_j is deterministic net surplus.

- (a) Derive the odds ratio $P(j \mid A)/P(k \mid A)$.
- (b) Prove the independence-of-irrelevant-alternatives property.
- (c) Explain precisely which part of IIA is an empirical restriction and which part is an algebraic implication of the model.

Problem 6.3 [Core]. A finite random-utility mixture.

There are alternatives a, b, c . A population consists of three strict preference types: $a \succ b \succ c$ with mass 0.5, $c \succ a \succ b$ with mass 0.2, and $b \succ a \succ c$ with mass 0.3.

- (a) Compute choice probabilities from menus $\{a, b\}$ and $\{a, b, c\}$.
- (b) Let the columns of M correspond to all six strict orders on $\{a, b, c\}$, with zero masses allowed. Stack the observed probabilities as $\mathbf{P}$ and write $\mathbf{P} = M\lambda$.
- (c) Explain why nonnegativity of λ and $\mathbf{1}_6^\top \lambda = 1$ are the central finite random-utility feasibility restrictions.

Part II: Formal and applied extensions

This chapter moves directly from the hand benchmark to its chapter use case.

Red bus, blue bus, and duplicated products

Chapter use case. Seed: Problems 6.1, 6.2. Fixed: all unlisted primitives and assumptions. Scaled: the number of observationally identical bus labels under independent-logit shocks. Tool: Codex + Python; Lean for the IIA core. Verification: Recover the one-bus and two-bus

probabilities, verify that probabilities sum to one, and separate the verified logit odds result from any shared-shock comparison model.

Problem 6.2 [Applied]. The red-bus blue-bus problem.

Walking and a red bus initially have systematic utilities normalized so that each has probability one half under logit. A blue bus with the same systematic utility as the red bus is added.

- (a) Compute the new logit probabilities.
- (b) Compare the total bus share before and after entry.
- (c) Explain why a model with highly correlated bus-specific shocks would predict a different substitution pattern.

Chapter 7

Limited Attention and Costly Information

Part I: By hand

Problem 7.1 [Core]. Sparse-max quasilinear demand.

Let $u(x_1, x_2) = \sqrt{x_1} + x_2$. The consumer chooses $x \in \mathbb{R}_+^2$, uses perceived prices $\hat{p} \gg 0$ in the first-order condition, and must satisfy the actual budget $p \cdot x = w$, where $p \gg 0$ and $w > 0$.

- (a) Derive the interior sparse-max choice.
- (b) State the condition under which the implied x_2 is nonnegative.
- (c) Show that ordinary Marshallian demand is recovered when $\hat{p} = p$, including the corner when the interior condition fails.

Problem 7.2 [Core]. CES demand under perceived prices.

Let $u(x) = [\alpha x_1^\rho + (1 - \alpha)x_2^\rho]^{1/\rho}$, where $0 < \alpha < 1$ and $\rho < 1$, $\rho \neq 0$. Consider an interior choice $x \in \mathbb{R}_{++}^2$ with perceived prices $\hat{p} \gg 0$, actual prices $p \gg 0$, and wealth $w > 0$.

- (a) Use perceived prices $\hat{p}$ to derive the chosen ratio x_1/x_2 .
- (b) Scale the bundle so that it exhausts the actual budget $p \cdot x = w$.
- (c) Identify separately the perceived-price composition effect of changing $\hat{p}$ and the feasibility effect of changing p . Explain why the first is not automatically a Hicksian substitution effect.

Part II: Formal and applied extensions

This chapter moves directly from the hand benchmark to its chapter use case.

Paying for information before choosing

Chapter use case. Seed: Problem 7.3. Fixed: all unlisted primitives and assumptions. Scaled: the common number N of equally likely states and matching actions, beginning at $N = 2$. Tool: Codex + Python. Verification: Recover the binary optimum, enforce conditional-probability and information-cost identities, and label any symmetry restriction used in the N -state extension. Problems 7.1 and 7.2 are comparison models, not derivations of rational inattention.

Problem 7.3 [Applied]. A binary rational-inattention problem.

There are two equally likely states and two actions. The decision maker receives payoff one from the correct action and zero from the incorrect action. A symmetric strategy is correct with probability $q \in [1/2, 1]$ and pays Shannon-information cost $\kappa[\log 2 - H(q)]$, where $H(q) = -q \log q - (1 - q) \log(1 - q)$ and $\kappa > 0$.

- (a) Write expected net payoff as a function of q .
- (b) Derive the interior optimum and show that $q^* = \exp(1/\kappa)/(1 + \exp(1/\kappa))$.
- (c) Describe the limits as $\kappa \downarrow 0$ and $\kappa \uparrow \infty$.

Part III

Aggregation and Markets

Chapter 8

Aggregate Demand and Representative Consumers

Part I: By hand

Problem 8.1 [Core]. Gorman aggregation.

Suppose consumer i has indirect utility $v_i(p, w_i) = a_i(p) + b(p)w_i$, with the same positive function b for every consumer. Assume a_i and b are differentiable on the relevant positive-price domain and that each indirect utility satisfies the regularity conditions for Roy's identity.

- (a) Apply Roy's identity to derive individual demand.
- (b) Show that aggregate demand depends on the wealth distribution only through total wealth $W = \sum_i w_i$.
- (c) Construct a representative indirect utility that generates aggregate demand.

Problem 8.2 [Core]. Aggregation under common homothetic preferences.

All consumers have the same homothetic preferences, strictly positive wealth $w_i > 0$, and a single-valued demand. Normalize the unit-wealth demand as $x(p, 1)$.

- (a) Show that $x_i(p, w_i) = w_i x(p, 1)$.
- (b) Derive aggregate demand and identify a representative consumer.
- (c) Explain why common homotheticity is sufficient but not necessary for exact aggregation.

Part II: Formal and applied extensions

This chapter moves directly from the hand benchmark to its chapter use case.

Aggregation when household wealth moves with prices

Chapter use case. Seed: Problems 8.1, 8.2. Fixed: all unlisted primitives and assumptions. Scaled: the number of consumers, with each wealth rule and preference class fixed. Tool: Codex + Python; Lean for the finite-sum core. Verification: Verify the finite aggregation identity and isolate every wealth-revaluation term. Do not describe the derivative or Slutsky bridge as Lean verified.

Problem 8.3 [Proof]. Why uncompensated aggregate price derivatives need not be symmetric.

Let $x^A(p) = \sum_i x_i(p, w_i(p))$, where individual wealth may vary with prices through endowments or profit shares.

- (a) Differentiate x^A with respect to prices.
- (b) Separate compensated substitution terms from wealth-revaluation terms.
- (c) Explain why symmetry and negative semidefiniteness of each individual Slutsky matrix do not generally imply the same properties for the uncompensated derivative $D_p x^A$.

Part IV

Production and Equilibrium

Chapter 9

Theory of the Firm

Part I: By hand

Problem 9.1 [Core]. Cobb-Douglas production and cost.

A firm has production function $f(\mathbf{z}) = z_1^\alpha z_2^\beta$ with $\alpha, \beta > 0$.

- (a) Classify returns to scale as a function of $\alpha + \beta$.
- (b) Derive conditional factor demands and the cost function for output $q > 0$ and input prices $\mathbf{w} \gg 0$.
- (c) Verify that cost is homogeneous of degree one and concave in $\mathbf{w}$.

Problem 9.3 [Core]. Constant returns and unbounded profit.

Assume Y is a cone and $0 \in Y$.

- (a) Prove that either $\pi_Y(p) = 0$ or $\pi_Y(p) = +\infty$.
- (b) Characterize the price vectors for which profit is zero.
- (c) Explain why a competitive constant-returns firm can operate at any scale only when the zero-profit condition holds.

Part II: Formal and applied extensions

Problem 9.2 [Proof]. Dual properties of the profit function.

Let $Y \subseteq \mathbb{R}^L$ be nonempty, closed, and convex. Assume that profit $\pi_Y(p) = \sup_{y \in Y} p \cdot y$ is finite on a neighborhood of the price under study and that the supremum is attained.

- (a) Prove that π_Y is convex and homogeneous of degree one in prices.
- (b) If supply is unique at p , use a subgradient argument to prove Hotelling's lemma and explain why the profit function is differentiable there.
- (c) State what changes when supply is set-valued.

Input substitution in a multi-input firm

Chapter use case. Seed: Problem 9.1. Fixed: all unlisted primitives and assumptions. Scaled: the input dimension from two to L , with the output target and technology family fixed. Tool: Codex + Python; Lean for finite algebra. Verification: Recover the two-input cost benchmark;

check Shephard derivatives, Hessian symmetry, negative semidefiniteness, and the homogeneity null direction; state the analytic assumptions not encoded in Lean.

Problem 9.4 [Proof]. Shephard's lemma and substitution.

Suppose the minimum in $C(\mathbf{w}, q) = \min\{\mathbf{w} \cdot \mathbf{z} : f(\mathbf{z}) \geq q, \mathbf{z} \geq 0\}$ exists, conditional factor demand $\mathbf{Z}(\mathbf{w}, q)$ is single-valued, and C is twice differentiable in $\mathbf{w}$.

- (a) Prove Shephard's lemma.
- (b) Show that $D_{\mathbf{w}}\mathbf{z} = D_{\mathbf{w}\mathbf{w}}^2 C$ is symmetric and negative semidefinite.
- (c) Use homogeneity to show $D_{\mathbf{w}}\mathbf{z}(\mathbf{w}, q)\mathbf{w} = 0$.

Chapter 10

Competitive Markets and Partial Equilibrium

Part I: By hand

Problem 10.1 [Core]. Linear partial equilibrium.

Demand and supply are $Q^d(p) = a - bp$ and $Q^s(p) = d(p - c)$ on their economically relevant ranges, with $a, b, d > 0$, $c \geq 0$, and $a > bc$.

- (a) Solve for equilibrium price and quantity.
- (b) Compute consumer and producer surplus using the choke price a/b of inverse demand and the price intercept c of inverse supply, so that the usual triangular-surplus geometry applies.
- (c) Verify that the competitive quantity maximizes total surplus.

Part II: Formal and applied extensions

Problem 10.2 [Applied]. A per-unit tax.

Introduce a per-unit tax $t > 0$ paid by sellers into the market of Problem 10.1, so buyers pay p_b and sellers receive $p_s = p_b - t$. Assume t is small enough that equilibrium quantity remains nonnegative.

- (a) Solve for p_b , p_s , and quantity.
- (b) Derive tax incidence. Express the local burden shares using the demand and supply slopes, and interpret them using point elasticities evaluated at the common no-tax equilibrium.
- (c) Compute deadweight loss and its derivative with respect to t .

Problem 10.3 [Proof]. Planner and competitive allocation.

A representative consumer has quasilinear utility $u(q) + m$, with enough transferable numeraire that the optimum is not constrained by a lower bound on m . The functions u and C are differentiable on $(0, \infty)$, have the required right derivatives at zero, u is increasing and strictly concave, and C is increasing and convex. Assume the surplus maximum over $q \geq 0$ is attained.

- (a) State and solve the planner's problem using Kuhn-Tucker conditions.
- (b) Show that any interior competitive equilibrium satisfies the same condition.
- (c) State the assumptions required for the converse decentralization claim.

Music sales and concerts

Chapter use case. Seed: Problems 1.2, 9.1, 10.1, 10.3. Fixed: all unlisted primitives and assumptions. Scaled: the music-volume or royalty share τ from the two benchmark values to a grid. Tool: Codex + Python. Verification: Reproduce the demand and cost formulas and both endpoint comparisons; bracket every stationary point, check both price boundaries, establish the global maximizer, and keep the volume-share and royalty-share scenarios distinct.

Problem 10.4 [Applied]. The Economics of Superstars: Music Sales and Concerts.

Taylor S. is a stylized superstar whose fans consume music services and concerts. This is the graduate version of the music-and-concerts case, with explicit preferences, technology, and market assumptions. All numbers are hypothetical; the model is not an empirical description of Taylor Swift's contracts. Quantities are divisible, and the mass of identical fans is one.

Part I. Fan demand

Write q_M, q_C for music and concert services, the two coordinates of consumption. A fan has wealth $w > 0$, prices $p_M, p_C > 0$, and utility $u(q_M, q_C) = (\frac{1}{2}q_M^\rho + \frac{1}{2}q_C^\rho)^{1/\rho}$, where $0 < \rho < 1$. The letters M, C are local commodity labels.

(a) Formulate the budget problem and derive demands using $\sigma = 1/(1 - \rho)$. For the remaining parts set $\rho = 1/2$, $w = 1$, and $p_M = 1$. Explain why music demand changes when the concert price changes.

Part II. Production

For $j \in \{M, C\}$ the artist's technology is $y_j = A_j \sqrt{n_j}$, where $n_j \geq 0$ is labor and $A_j > 0$ productivity. Labor costs $\nu > 0$ per hour, distinct from fan wealth w . The processes are independent. This stylized music-service cost does not describe the cost of creating an original recording as a function of its streams.

(b) Derive $C_j(\nu, y_j)$, average cost, and marginal cost. Explain decreasing returns. Set $A_M = A_C = \nu = 1$ in the numerical parts.

Part III. A share of volume and receipts

An exogenous $\tau \in (0, 1]$ is the fraction of music consumption attributable to this artist. Her output is $y_M = \tau q_M$ and her music receipts are $\tau p_M q_M$. Concert output is $y_C = q_C$. She takes p_M, τ as given and chooses $p_C > 0$. There is no shared labor constraint or fixed cost.

(c) Write total revenue and profit after substituting demand and costs. Distinguish a revenue derivative with respect to ticket price from marginal revenue with respect to output. Account for changes in both revenue streams and production costs.

(d) Derive the first-order condition for the optimal concert price. Examine profit as $p_C \downarrow 0$ and $p_C \rightarrow \infty$. Establish global optimality and uniqueness, analytically or with a justified numerical method accounting for every stationary point and boundary.

(e) Compare $\tau = 1$ and $\tau = 1/2$. In a table report optimal concert prices, both fan quantities, both delivered outputs, revenues, costs, and profits. State numerical accuracy. Does the concert price rise or fall when the music share falls?

(f) Use the first-order condition or implicit differentiation to analyze the price response for $\tau \in [1/2, 1]$. Interpret the roles of substitution and production costs. Do not assume the same sign for every technology or parameter value.

Part IV. A royalty share

Now τ changes only the fraction of music revenue paid to the artist: music output is $y_M = q_M$, while receipts remain $\tau p_M q_M$. All other assumptions are unchanged.

(g) Write the revised profit function and repeat the comparison between $\tau = 1$ and $\tau = 1/2$. Identify the changed cost channel and determine whether the earlier price response survives.

(h) Explain what evidence distinguishes a fall in music volume from a fall in the royalty fraction. Discuss the roles of CES substitution, decreasing returns, and control of concert prices. Contrast the artist's decision with price-taking supply in the main chapter.

Chapter 11

General Equilibrium Theory

Part I: By hand

Problem 11.1 [Core]. Cobb-Douglas exchange equilibrium.

There are two consumers with common utility $u_i(x_{i1}, x_{i2}) = x_{i1}^\alpha x_{i2}^{1-\alpha}$, $0 < \alpha < 1$, and endowments $\omega_1 = (1, 2)$, $\omega_2 = (2, 1)$.

- (a) Derive individual Walrasian demands.
- (b) Normalize $p_2 = 1$ and solve for the equilibrium relative price.
- (c) Compute equilibrium allocations and verify both market-clearing equations.

Part II: Formal and applied extensions

Problem 11.2 [Proof]. First Fundamental Welfare Theorem.

Consider a private-ownership economy with locally nonsatiated preferences and profit-maximizing firms.

- (a) State the definition of Pareto efficiency used in the theorem.
- (b) Prove that a competitive equilibrium allocation is Pareto efficient.
- (c) Identify exactly where local nonsatiation, budget feasibility, profit maximization, and market clearing enter the proof.

Problem 11.3 [Proof]. Supporting a Pareto optimum.

Suppose consumption sets and production sets are closed and convex, preferences are continuous, convex, and locally nonsatiated, and a feasible allocation is Pareto efficient. Assume free disposal and that the aggregate net-resource-requirement set used below is closed, convex, and upward comprehensive. Assume the remaining separation hypotheses stated in the chapter.

- (a) Formulate the aggregate set-separation argument and show how it supports the allocation as a price quasi-equilibrium with transfers.
- (b) Explain why the initial endowment distribution generally must be changed by lump-sum transfers.
- (c) State the positive-wealth or boundary condition needed to upgrade the quasi-equilibrium to a price equilibrium with transfers, and distinguish that object from a competitive equilibrium for the original ownership economy.

A consumer-owned firm

Chapter use case. Seed: Problems 11.1, 11.4. Fixed: all unlisted primitives and assumptions. Scaled: the production exponent from one-half to $\beta \in (0, 1)$. Tool: Codex + Python. Verification: Recover the square-root economy; verify planner and decentralized allocations coincide; check second-order conditions, profit finiteness, both market-clearing equations, and limiting behavior in β .

Problem 11.4 [Applied]. One consumer and one producer.

There are two goods. Good 1 can be consumed or used as input z to produce good 2 according to $y_2 = \sqrt{z}$. The consumer owns the firm and has an endowment $\bar{\omega}_1 > 0$ of good 1 and no good 2. Utility is $\log x_1 + \log x_2$.

- (a) Solve the planner's problem.
- (b) Find supporting relative prices from marginal conditions.
- (c) Verify that the consumer and firm choices clear both markets.

Chapter 12

Positive Theory of Equilibrium

Part I: By hand

Problem 12.1 [Core]. Differential identities for excess demand.

Let aggregate excess demand $z : \mathbb{R}_{++}^L \rightarrow \mathbb{R}^L$ be continuously differentiable, homogeneous of degree zero, and satisfy Walras' law.

- (a) Differentiate homogeneity to show $D_p z(p)p = 0$.
- (b) Differentiate Walras' law to show $[D_p z(p)]^\top p = -z(p)$.
- (c) Specialize the second identity to an equilibrium and interpret the left null vector.

Part II: Formal and applied extensions

Problem 12.2 [Proof]. Regularity and local uniqueness.

Normalize prices by $p_L = 1$ and let $\hat{z}$ be the first $L - 1$ excess-demand equations.

- (a) Define a regular equilibrium.
- (b) Use the inverse function theorem to prove that a regular equilibrium is locally isolated.
- (c) Suppose every normalized equilibrium is regular. Explain why compactness and boundary conditions then imply that there are only finitely many equilibria.

Problem 12.3 [Proof]. Gross substitutes and uniqueness.

Assume aggregate excess demand satisfies strict gross substitutes on the relevant price domain: increasing one price strictly raises excess demand for every other good whenever the comparison is nondegenerate. Suppose two strictly positive normalized equilibrium price vectors exist.

- (a) Compare the largest relative price change across goods.
- (b) Use gross substitutes and Walras' law to derive a contradiction unless the two price vectors are proportional.
- (c) Explain why weak gross substitutes alone need not imply uniqueness and where strictness (or an irreducibility substitute) enters the proof.

A coalition challenges a competitive allocation

Chapter use case. Seed: Problem 12.4. Fixed: all unlisted primitives and assumptions. Scaled: the blocking predicate from all-strict improvement to weak improvement for all members and strict improvement for one witness. Tool: Codex + Lean. Verification: Compile the Lean strong-coalition price-support declaration and match each hypothesis to the economic proof. Lean verifies the finite price-summing contradiction, not the derivation of price support from Walrasian optimization or the replica-economy convergence claim. This case closes the price-support and core thread; Problems 12.2 and 12.3 are separate regularity and uniqueness extensions.

Problem 12.4 [Core]. Competitive allocations and the strong core.

Use the strong-blocking convention in a finite pure-exchange economy with locally nonsatiated preferences. A nonempty coalition S strongly blocks x when it can choose $(y_i)_{i \in S}$ such that $\sum_{i \in S} y_i = \sum_{i \in S} \omega_i$, $y_i \succeq_i x_i$ for every $i \in S$, and $y_j \succ_j x_j$ for at least one $j \in S$. Define the associated strong core.

- (a) Prove that every Walrasian allocation belongs to the strong core.
- (b) Give an economic reason why the converse can fail with finitely many consumers.
- (c) State the content, but not a proof, of the core-convergence theorem for replica economies.

Part V

**Advanced Equilibrium
Applications**

Chapter 13

Testable Restrictions on the Equilibrium Manifold

Part I: By hand

Problem 13.1 [Core]. Equilibrium Afriat feasibility.

At each observation t , prices $p^t \gg 0$, individual incomes $(w_i^t)_i$, and aggregate endowment ω^t are observed, while individual allocations are latent.

(a) Introduce latent consumptions x_i^t and write individual budget equalities $p^t \cdot x_i^t = w_i^t$ and aggregate market clearing $\sum_i x_i^t = \omega^t$.

(b) Write Afriat inequalities for every consumer across observations.

(c) Explain why joint feasibility is a finite semialgebraic revealed-equilibrium test, why the latent multipliers and bundles make the full system bilinear, and how quantifier elimination projects the restrictions onto the observables.

Part II: Formal and applied extensions

Problem 13.2 [Proof]. What equilibrium data identify.

Suppose only equilibrium prices and aggregate endowments are observed.

(a) List the preference, allocation, and ownership objects that remain unobserved.

(b) Explain why market clearing alone cannot identify individual demand.

(c) Describe how additional observations, known individual endowments, or parametric restrictions can turn an underidentified model into a testable one.

A counterfactual equilibrium identified set

Chapter use case. Seed: Problems 13.1, 13.3. Fixed: all unlisted primitives and assumptions. Scaled: the number of equilibrium observations from two to T , with the maintained preference class and counterfactual fixed. Tool: Codex + Python. Verification: Verify feasibility and market clearing for every retained candidate, show how each added observation changes the bounds, and never replace set identification by a point estimate.

Problem 13.3 [Applied]. Counterfactual discipline.

A dataset passes a revealed-equilibrium feasibility test under a maintained class of preferences.

- (a) Explain why passing the test does not point-identify a counterfactual equilibrium.
- (b) Define an identified set for a counterfactual price or allocation.
- (c) State two economically meaningful restrictions that could tighten this set, and explain the cost of each restriction.

Chapter 14

Sequential Trade

Part I: By hand

Problem 14.1 [Core]. Spanning contingent consumption.

There are S states tomorrow and K traded assets with payoff matrix $R \in \mathbb{R}^{S \times K}$ and date-zero asset-price vector $q \in \mathbb{R}^K$. Portfolios are unrestricted, $\theta \in \mathbb{R}^K$, so short sales are allowed. Assume no arbitrage.

(a) State the rank condition under which every contingent net trade can be implemented by an asset portfolio.

(b) If $S = 2$ and the assets have payoff vectors $(1, 1)$ and $(1, 0)$, construct the portfolio that delivers any target payoff (c_1, c_2) .

(c) Let $\psi \in \mathbb{R}_{++}^S$ be state prices satisfying $q = R^\top \psi$. Explain how the date-zero portfolio budget maps into an Arrow-Debreu present-value budget.

Problem 14.2 [Core]. Incomplete markets.

Let $S = 3$ and suppose the only assets are a bond with payoff $(1, 1, 1)$ and an asset with payoff $(1, 0, 0)$.

(a) Characterize the attainable payoff subspace.

(b) Determine whether the claim $(0, 1, -1)$ is attainable.

(c) Explain why competitive equilibrium with these assets need not be Pareto efficient relative to the full contingent-commodity economy.

Part II: Formal and applied extensions

This chapter moves directly from the hand benchmark to its chapter use case.

Complete versus incomplete sequential markets

Chapter use case. Seed: Problems 14.1, 14.2. Fixed: all unlisted primitives and assumptions. Scaled: the asset span from rank $S - 1$ to rank S at a fixed number of states. Tool: Codex + Python; Lean target. Verification: Check portfolio replication, the state-price units map, and

present-value budgets; include a rank-deficient claim that cannot be implemented. Arrow-Radner equivalence remains a future Lean target.

Problem 14.3 [Proof]. Arrow-Debreu and sequential equivalence.

There are S states, K assets with payoff matrix $R \in \mathbb{R}^{S \times K}$ of rank S , date-zero asset prices q , and state- s spot-price vectors p_s . Asset payoffs are denominated in the state numeraire. Consumer i chooses a portfolio z_i subject to $q \cdot z_i \leq 0$, then chooses x_{is} subject to $p_s \cdot x_{is} \leq p_s \cdot \omega_{is} + p_s 1 (Rz_i)_s$. Assume no arbitrage and strictly monotone continuous preferences.

- (a) Map an Arrow-Debreu budget into date-zero asset purchases and state-by-state spot trades.
- (b) Prove the reverse mapping from a sequential budget to a present-value budget.
- (c) Identify which step fails under incomplete markets or binding portfolio constraints.

Chapter 15

Matching

Part I: By hand

Problem 15.1 [Core]. Deferred acceptance by hand.

Three men have rankings $m_1 : w_1 \succ w_2 \succ w_3$, $m_2 : w_1 \succ w_2 \succ w_3$, and $m_3 : w_2 \succ w_3 \succ w_1$. The women rank $w_1 : m_2 \succ m_1 \succ m_3$, $w_2 : m_1 \succ m_3 \succ m_2$, and $w_3 : m_3 \succ m_2 \succ m_1$.

- (a) Run men-proposing deferred acceptance round by round.
- (b) Verify stability of the resulting matching by checking potential blocking pairs.
- (c) Explain why termination is guaranteed in every finite market.

Part II: Formal and applied extensions

Problem 15.2 [Proof]. Proposer optimality.

Consider a finite balanced one-to-one matching market in which every participant has a complete strict ranking of the other side and all partners are acceptable.

- (a) Prove that if a receiver rejects a proposer during deferred acceptance, that rejected proposer-receiver pair cannot be matched to each other in any stable matching.
- (b) Use this rejection lemma to prove proposer optimality among stable matchings.
- (c) Explain why the receiver side obtains its least-preferred stable matching.

Housing allocation and coalitional stability

Chapter use case. Seed: Problem 15.3. Fixed: all unlisted primitives and assumptions. Scaled: the number of agents and houses from three to n . Tool: Codex + Python; Lean target. Verification: Recover the three-agent top-trading-cycles allocation; check permutation, individual rationality, Pareto comparisons, and coalition blocking by enumeration. Top trading cycles and housing-core claims remain future Lean targets.

Problem 15.3 [Core]. Top trading cycles.

Agents 1, 2, 3 initially own houses h_1, h_2, h_3 . Their rankings are $1 : h_2 \succ h_1 \succ h_3$, $2 : h_1 \succ h_2 \succ h_3$, and $3 : h_1 \succ h_3 \succ h_2$.

- (a) Run top trading cycles.

- (b) Show that the outcome is individually rational and Pareto efficient.
- (c) Verify directly that no coalition blocks the outcome.

Chapter 16

Computable General Equilibrium

Part I: By hand

Problem 16.1 [Core]. Calibrating a CES technology.

A constant-returns technology is $Q = A[\alpha K^{-\rho} + (1 - \alpha)n^{-\rho}]^{-1/\rho}$, with $\rho > -1$ and $\rho \neq 0$. A strictly positive interior cost-minimizing benchmark observation gives (Q, K, n, r, W) and an elasticity of substitution $\sigma > 0$, $\sigma \neq 1$. (The Cobb-Douglas limit $\sigma = 1$ is outside the displayed parameterization.)

- (a) Recover ρ from σ .
- (b) Use the cost-minimizing first-order condition to recover α .
- (c) Recover A and verify that the benchmark is exactly replicated.

Problem 16.2 [Core]. Social-accounting consistency.

A two-sector social accounting matrix records sectoral sales, intermediate purchases, labor and capital payments, household income, taxes, savings, and imports.

- (a) State the row-balance and column-balance conditions.
- (b) Explain why every market-clearing equation has a corresponding accounting identity but not every accounting identity is a behavioral equation.
- (c) Design a numerical residual check that would detect an incorrectly calibrated benchmark.

Part II: Formal and applied extensions

This chapter moves directly from the hand benchmark to its chapter use case.

A tariff counterfactual from partial to general equilibrium

Chapter use case. Seed: Problems 16.1, 16.2, 16.3. Fixed: all unlisted primitives and assumptions. Scaled: model closure from fixed factor prices to one endogenous shared labor market linking import-competing and export sectors. Tool: Codex + Python. Verification: Replicate the partial-equilibrium benchmark, then report factor-market, goods-market, trade-balance, government-budget, and welfare residuals. Identify which change comes from the single added general-equilibrium channel.

Problem 16.3 [Applied]. A tariff counterfactual.

A small open economy takes the world price P^W and exchange rate ER as given. A tariff τ implies domestic price $P = \text{ER } P^W (1 + \tau)$. Let differentiable domestic demand and supply be $D(P)$ and $S(P)$, imports be $M(P) = D(P) - S(P) > 0$, and tariff revenue be returned lump-sum, $\text{TR}(\tau) = \tau \text{ER } P^W M(P(\tau))$. Interpret D as quasi-linear (or otherwise free of income effects) and S as competitive supply generated by marginal cost, so consumer and producer surplus are valid partial-equilibrium welfare measures.

- (a) Derive the first-order change in the domestic price from a small tariff change.
- (b) Derive the symbolic first-order decomposition $d\mathcal{W} = d\text{CS} + d\text{PS} + d\text{TR}$ and simplify it at free trade. Do not assume numerical functional forms.
- (c) Explain which general-equilibrium channels are omitted by that approximation and must be restored in a CGE counterfactual.

Appendix A

Order Theory

Part I: By hand

Problem A.1 [Core]. Classifying relations.

Assume $|X| \geq 2$. For each relation below, determine whether it is reflexive, irreflexive, symmetric, asymmetric, antisymmetric, transitive, and complete: equality on X ; strict set inclusion on 2^X ; weak set inclusion on 2^X ; and the componentwise order on $\mathbb{R}^2$.

Give a counterexample for every failed property.

Part II: Formal and applied extensions

Problem A.2 [Proof]. Quotienting a preorder.

Let R be a preorder on X and define xI_Ry if and only if xRy and yRx .

- (a) Prove that I_R is an equivalence relation.
- (b) Define $[x]\bar{R}[y]$ if and only if xRy . Prove that this definition is independent of representatives.
- (c) Prove that $\bar{R}$ is a partial order on X/I_R .

Problem A.3 [Proof]. Maximal, greatest, and finite existence.

Let R be a preorder on a nonempty finite set S .

- (a) Prove that S has an R -maximal element.
- (b) Give an example with several maximal elements and no greatest element.
- (c) Prove that if R is complete, maximal and greatest elements coincide.

Finite preference completion and multiple utilities

Chapter use case. Seed: Problems A.1, A.2, A.3. Fixed: all unlisted primitives and assumptions. Scaled: the number $|X|$ of alternatives in a finite preorder, beginning with the hand example. Tool: Codex + Lean. Verification: Verify the preorder, quotient construction, and each proposed linear extension. Szpilrajn extension is currently verified; quotient pullback, intersection of all extensions, and the finite multiple-utility corollary remain explicit formal targets.

Problem A.4 [Proof]. Order extensions and multiple representation.

Let R be a preorder on a finite set X .

- (a) Pass to the quotient partial order on X/I_R and extend it to a linear order.
- (b) Pull the extension back to a complete preorder on X .
- (c) Show that R is the intersection of its complete preorder extensions and deduce a finite multiple-utility representation.

Additional Practice Reserve

These ten question-only problems are corrected and independently checked descendants of earlier course assessments. Each combines a hand-solvable benchmark with a controlled computational or formal scale-up.

A.1 Revealed Preference and Recoverability

Problem R001 [Proof]. Quasilinear Afriat inequalities and welfare bounds.

Suggested weekly placement: PS03.

Let $\mathcal{O} = \{(p^t, x^t)\}_{t=1}^T$, where $p^t \gg 0$ and $x^t \in \mathbb{R}_+^L$.

Part I -- by hand. Prove that the observations can satisfy $x^t \in \arg \max_{x \in \mathbb{R}_+^L} \{u(x) - p^t \cdot x\}$ only if there are numbers $v^1, \dots, v^T$ such that $v^t - v^s \geq p^t \cdot (x^t - x^s)$ for every s, t . Prove the converse by studying $u(x) = \min_t \{v^t + p^t \cdot (x - x^t)\}$. Establish continuity, concavity, strict monotonicity, and rationalization rather than merely citing Afriat's theorem.

For $V(p) = \max_{x \in \mathbb{R}_+^L} \{u(x) - p \cdot x\}$, suppose optimizers $x(p)$ and $x(q)$ exist. Derive upper and lower bounds on $V(p) - V(q)$ using the two observed demands. Extend the argument to a finite path of price vectors and state precisely when a componentwise price increase gives a strict welfare loss.

Part II -- verified scale-up. Express the inequalities as a difference-constraints graph. With Codex and Python, implement a feasibility test for arbitrary T and L , return one normalized vector v , and report a violating positive cycle when the system is infeasible. Unit-test the code on a small rationalizable data set and on a perturbed data set. Verify every returned inequality directly. As an optional Lean task, formalize the telescoping path bound conditional on the finite system of inequalities.

A.2 Abstract Choice and Order Foundations

Problem R002 [Proof]. Sequential satisficing with a default rule.

Suggested weekly placement: PS04.

Let X be finite, let $\triangleright$ be a strict linear (total) search order, and let $S = \{x : u(x) \geq \bar{u}\}$. From a nonempty menu A , inspect alternatives in the order $\triangleright$ and choose the first member of $A \cap S$. If $A \cap S$ is empty, choose the $\triangleright$ -first member of A .

Part I -- by hand. Construct a single strict priority order $\succ^*$ whose maximal element agrees with this procedure on every nonempty menu. Use that representation to prove WARP

and SARP. Explain exactly why the fallback clause is necessary for a full-domain choice function. Give a three-alternative example and list the choice from every nonempty menu.

Part II -- verified scale-up. Encode finite menus, the search order, the threshold set, and the fallback choice. Ask Codex to generate either a Lean proof or an exhaustive checker establishing equality between sequential choice and $\succ^*$ -maximization. Scale from three alternatives to all menus of a user-chosen finite set, and include a deliberately broken implementation without the fallback to exhibit its failure of full-domain nonemptiness.

Problem R003 [Applied]. Choice overload and attraction through menu-dependent attention.

Suggested weekly placement: PS06.

An outside option o is always considered. In a menu with n active products, each active product is independently considered with baseline probability $q_n = 1/n^2$; the consumer chooses the highest- u considered item, or o if none is considered. Utilities are strict. When a target b and its dominated decoy b^- are both offered, the consideration probability of b rises by $\delta = 1/3$, while all other baseline probabilities remain unchanged.

Part I -- by hand. Compare $A_2 = \{a, c, o\}$ with $A_3 = \{a, c, d, o\}$ and calculate the probability of choosing o . Next let $u(b) > u(c) > u(b^-) > u(o)$ and compare $A = \{b, c, o\}$ with $B = \{b, b^-, c, o\}$. Calculate the probability of choosing b in both menus. Prove that the resulting attraction inequality cannot be generated by any menu-independent random-utility model.

Part II -- verified scale-up. Build an exact enumerator over consideration sets and a Monte Carlo simulation. Confirm the hand calculations, then vary n , δ , and the ranking. Ask Codex to produce automated checks for probability adding-up, the parameter domain, regularity violations, and convergence of the simulation to the exact probabilities. Report which changes generate overload, attraction, both, or neither.

Problem R004 [Proof]. Maximal choice under incompleteness and a GARP failure.

Suggested weekly placement: PS04.

Let $X = \{a, b, c\}$. Define an incomplete preorder R by reflexivity together with bPa ; alternative c is incomparable with both a and b . From each nonempty menu A , let $C(A)$ contain all R -maximal elements.

Part I -- by hand. Compute $C(A)$ for every nonempty menu. Define direct revealed preference by xR^Dy when some menu chooses x while making y feasible, and define xP^Dy when that menu rejects y . Show that the transitive closure $(R^D)^*$, together with P^D , violates the standard GARP test; identify an explicit witness. Explain why this does not contradict maximality under an incomplete preorder.

Part II -- verified scale-up. Represent finite preorders and maximal-choice correspondences in Lean or Python. Verify reflexivity and transitivity of the primitive relation, compute all maximal sets, and mechanically exhibit the revealed-preference path. Then enumerate all three-alternative preorders and classify which induced maximal-choice data satisfy GARP. Distinguish a failure of the data test from a failure of the primitive preorder axioms.

A.3 Production and Competitive Markets

Problem R005 [Proof]. Aggregating heterogeneous Cobb--Douglas firms.

Suggested weekly placement: PS08.

Firm j has technology $q_j = A_j \ell_j^{a_j} k_j^{b_j}$ with $A_j > 0$, $a_j, b_j > 0$, and $a_j + b_j < 1$. Its net-output set is $Y_j = \{(-\ell, -k, q) : q \leq A_j \ell^{a_j} k^{b_j}, \ell, k \geq 0\}$. Output price is $p > 0$ and input prices are $w_\ell, w_k > 0$.

Part I -- by hand. Derive the unique profit-maximizing output, both input demands, and profit. State where decreasing returns are used. Prove that the aggregate technology is the Minkowski sum $Y = Y_1 + \dots + Y_J$ and that its profit function is the sum of the firm profit functions. State the corresponding Hotelling result both at differentiable prices and in subdifferential form, using derivatives $(q_j, -\ell_j, -k_j)$ with respect to (p, w_ℓ, w_k) .

Part II -- verified scale-up. Generate a heterogeneous population of firms and compute aggregate supply and factor demands. Verify numerically that direct maximization over firm-level allocations agrees with the sum of closed-form firm expressions. Use automatic or symbolic differentiation to check Hotelling's lemma, and test how the aggregate elasticity changes with the distribution of returns-to-scale parameters.

Problem R006 [Applied]. Selection versus capacity destruction in partial equilibrium.

Suggested weekly placement: PS09.

There is a mass $\eta > 0$ of consumers with $u(x, m; \alpha) = 2\sqrt{\alpha x} + m$ and $\alpha \sim U[0, 1]$. The numeraire is unrestricted, or wealth is sufficient for the interior solution. There is initially a unit mass of firms with $C(q; e) = q^2/(2e)$, where productivity has the uniform law on $e \in (0, 1)$; endpoint conventions are immaterial for aggregation.

Part I -- by hand. Derive individual and aggregate demand, firm supply and profit, and the competitive equilibrium price. Verify Hotelling's lemma. Compare two shocks: (i) the unit mass of firms is replaced by a unit mass with $e \sim U[1/2, 1]$; (ii) firms with $e < 1/2$ disappear without replacement. Solve for both new prices and explain why selection and capacity destruction have opposite price effects. At the baseline equilibrium, compute the cross-sectional variance of consumption and mean indirect utility when numeraire wealth is constant.

Part II -- verified scale-up. Replace the uniform type distributions by beta distributions and allow an arbitrary survival cutoff. Derive the market-clearing equation, solve it numerically over a parameter grid, and separate the effect of changing the conditional productivity distribution from changing producer mass. Include monotonicity checks and compare numerical integration with the uniform closed form.

A.4 Equilibrium, Trade, and Computation

Problem R007 [Applied]. CES structural gravity and multilateral resistance.

Suggested weekly placement: PS11.

Destination j has CES utility $U_j = [\sum_i a_i^{1/\sigma} c_{ij}^{(\sigma-1)/\sigma}]^{\sigma/(\sigma-1)}$ with $\sigma > 1$ and $a_i, p_i, \tau_{ij}, y_j > 0$. Delivered prices are $p_{ij} = p_i \tau_{ij}$ and bilateral expenditure is $x_{ij} := p_i \tau_{ij} c_{ij}$. Total income is $Y = \sum_j y_j$, and balanced expenditure and income impose market clearing $y_i = \sum_j x_{ij}$.

Part I -- by hand. Derive the CES price index, bilateral quantities, and bilateral expenditure. With $\theta_j = y_j/Y$, define an outward multilateral-resistance index and derive the structural-gravity equation together with the inward-index system. Take logs and identify the exporter effect, importer effect, and trade-cost coefficient. Explain the reciprocal scale freedom and state a normalization; do not claim that symmetric trade costs alone pin down every unnormalized inward and outward index as equal.

Part II -- verified scale-up. For a synthetic economy with at least ten countries, solve the normalized multilateral-resistance system in log variables. Verify market clearing, bilateral adding-up, positivity, and invariance to the discarded scale normalization. Compare fixed-point iteration with a root solver and report convergence diagnostics. Recover the trade-cost coefficient from a regression with exporter and importer fixed effects.

Problem R008 [Proof]. Why a Banach argument fails even when existence and uniqueness hold.

Suggested weekly placement: PS10.

On $K = [1/2, 2]^2$, consider $g : K \rightarrow \mathbb{R}^2$ with identical components $g_j(P) = [(P_1^{-1/2} + P_2^{-1/2})/2]^2$.

Part I -- by hand. Prove that g is a continuous self-map of K and use the appropriate fixed-point theorem to establish existence. Compute $Dg(1, 1)$, its eigenvalues, and its Euclidean operator norm. Explain why no contraction proof in any induced norm can follow from the stated map. Nevertheless, prove directly that the positive fixed point is unique. Exhibit a two-cycle of the un-damped Picard iteration.

Part II -- verified scale-up. Use interval arithmetic or Lean-supported inequalities to certify invariance of K and the derivative calculation. Compare plain iteration, the damped rule $P^{n+1} = (1 - \lambda)P^n + \lambda g(P^n)$ for $\lambda = 1/2$, and a root solver from multiple starting values. Ask Codex to create tests that distinguish existence, uniqueness, local stability, and global contraction; the report must not infer one property from another.

Problem R009 [Proof]. A regular two-good economy with three equilibria.

Suggested weekly placement: PS10.

Normalize prices to $(q, 1)$. Consumer 1 chooses $a \geq 0, b > 0$, has $u_1(a, b) = a - \frac{1}{8}b^{-8}$, and owns $(2, r)$. Consumer 2 chooses $c > 0, d \geq 0$, has $u_2(c, d) = -\frac{1}{8}c^{-8} + d$, and owns $(r, 2)$, where $r = 2^{8/9} - 2^{1/9}$.

Part I -- by hand. Derive both consumers' interior demands and prove that $a(q), d(q) > 0$ for every $q > 0$, so boundary demand cannot create an omitted equilibrium. Derive the good-2 excess-demand function. Show that $q \in \{1/2, 1, 2\}$ are equilibria and that there are no other positive normalized equilibria. Verify that each equilibrium is regular and locally isolated. Compute $z'_2(1)$ and compare it with the gross-substitutes requirement $z'_2(1) \geq 0$. Explain why regularity does not imply uniqueness.

Part II -- verified scale-up. Put $t = q^{1/9}$ and use a computer algebra system to certify the positive-root count without relying only on a plot. Check all equilibrium residuals and derivatives at high precision. Choose one scalar endowment perturbation, initialize continuation at each baseline equilibrium, and trace each local branch only until regularity fails or the declared small parameter interval ends. Connect the local numerical result to the implicit-function theorem. Optionally formalize the exact three roots and their nonzero derivatives in Lean.

Problem R010 [Applied]. A complete-markets Radner equilibrium by hand.

Suggested weekly placement: PS11.

There are two consumers, two states A, B , and two goods in each state. Both consumers have equal state probabilities and state utility $\log c_1 + \log c_2$. Each owns $\omega > 0$ units of good 2 in each state. Consumer 1's good-1 endowment is $(\omega/2, 3\omega/2)$ across states, and consumer 2's is reversed. Good 1 is the state numeraire and r_s is the state- s price of good 2. Arrow security s pays one unit of good 1 in state s . Consumer i holds θ_{is} , securities are in zero net supply, and the date-0 asset budget is $Q_A\theta_{iA} + Q_B\theta_{iB} = 0$.

Part I -- by hand. Normalize the state- B security price to $Q_B = 1$ and write $q = Q_A$. Derive individual security demands as functions of q, r_A, r_B . Show that commodity clearing under equal spot prices requires $r_A = r_B = 1$ and that the equilibrium security-price ratio is one. Calculate portfolios and state consumption, and verify both commodity and security clearing. Explain the role of strict concavity and price normalization in the uniqueness statement.

Part II -- verified scale-up. Generalize the computation to S states and a full set of Arrow securities. Build the sequential budget matrix, solve the equilibrium numerically, and verify state-by-state feasibility and zero net security supply. Ask Codex or Lean to check the linear identity mapping sequential budgets into the Arrow--Debreu present-value budget, clearly separating the algebraic equivalence from equilibrium existence.